Co-4

Message Digest

* Known as a hash value or hash code is a fixed-size numerical representation derived from a given input data of arbitrary length.

Properties

→ Deterministic = For the same input, a hash function will always produce the same digest.

→ Fixed size

→ Collision Resistance

MD5

128 bits	Digest Length
512 bits	Basic unit of processing
64 (4 rounds of 16)	No of steps
∞	Maximum message size
4	Primitive logical functions
64	Additive constants used
Little-endian	Endianess.

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Message Authentication Code (MAC)

* MAC function accepts a variable length message and a secret key as input and produces a fixed-length MAC value as output

$$MAC = MAC(K, M)$$

M = input message
K = shared secret key
C = MAC function

HMAC (Hash-Based Message Authentication Code)

* Uses a cryptographic hash function and a secret key to verify the integrity & authenticity of a message.

Steps

1) Sender & receiver share a secret key K.
2) The key is combined with the inner padding (ipad).
3) The result is combined with the message M & hashed.
4) The output is combined with the outer padding & key.

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5) A final Hash is generated as the HMAC value.

6) Receiver calculates HMAC again & compares both values.

$HMAC(K,m) = H((K' \oplus opad) \| H((K' \oplus ipad) \| m))$

$\|$ = Concatenation.

DAA (Digital Authentication Algorithm)

* This uses the DES encryption to generate a message Authentication Code (MAC).

Step 1:- Divide the message

* Divide the original message M into multiple blocks of 64 bits.

Step 2:- Process the first block

$O_1 = DES_K(D_1)$ ← input to the DES encryption algorithm.

↓ Secret key.

Step 3:- Process the second block

$O_2 = DES_{O_1}(D_2)$

Step 4:- Continue processing

$O_3 = DES_{O_2}(D_3)$

* Continue until the last block D_N.

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Step5:- Generate MAC.

* Take the leftmost 16 to 64 bits of the final output O_N.

MAC = Leftmost 16-64 bits (O_N)

CMAC (Cipher-Based Message Authentication Code)

* This uses a block cipher, such as AES and a secret key to verify message integrity & authenticity.

$$T = CMAC(K, m)$$

K = Secret key.
m = Message
T = Authentication tag.

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